## CS-111-TH: Problem Solving using Computer and C Programming
## Lectures: 30 (Credits-2)

**Prerequisites:** None

**Course Objectives:**

1. To introduce the foundations of computing, programming and problem- solving using computers.
2. To develop the ability to analyze a problem and devise an algorithm to solve it.
3. To develop the basic concepts and terminology of programming in general.
4. To implement algorithms in C.
5. To test, debug and execute programs.

**Learning Outcomes: On completion of this course, students will be able to:**

1. Explore algorithmic approaches to problem solving.
2. Develop modular programs using control structures and arrays in C.

**Unit 1: Problem solving and C programming** **08**

1.1 Problem solving using Computers
1.2 Programming Languages as tools, types of languages
1.3 Algorithms-definition, characteristics, examples, advantages and limitations
1.4 Flow charts - definition, notations, examples, advantages and limitations, Comparison with algorithms
1.5 Structure and example of first C program
1.6 Compilation process (compilers, interpreters)
1.7 Character set, Keywords, Identifiers
1.8 Variables, Constants (character, integer, float, string, escape sequences, enumeration, constant)
1.9 Data Types (Built-in and user defined data types)
1.10 Operators, Expressions, Types of operators
1.11 Arithmetic operators, Increment Decrement operators, Relational and logical operators, Bitwise operators, Assignment operators, Comma operator, size of operator, conditional operator, Operator precedence and Order of evaluation
1.12 Formatted input and output, format specifiers

**Unit 2: Control Structures** **07**

2.1 Decision making structures: if, if-else, else-if ladder, switch
2.2 Loop control structures - while, do while, for
2.3 Use of break and continue
2.4 Nested control structures
2.5 Unconditional branching (goto statement)

**Unit 3: Functions** **07**

3.1 Function definition, Types of functions (Standard library and User defined functions)
3.2 Function parameters/arguments (Actual, Formal)
3.3 Parameter passing method (by value), return statement
3.4 Recursive functions
3.5 Scope of variables and Storage classes

**Unit 4: Arrays** **08**

4.1 Concept of array, advantages, disadvantages
4.2 Types of Arrays – One, Two dimensional array
4.3 Array Operations - declaration, initialization, accessing array elements
4.4 Memory representation of two-dimensional array (row major and column major)
4.5 Passing arrays to function
4.6 Array applications - Linear search, sorting an array (bubble sort)

**Reference Books:**


1. A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India
2. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill
3. Programming in ANSI C, Ram Kumar and Rakesh Agrawal
4. The 'C' programming language, Brian Kernighan, Dennis Ritchie, PHI

# CS-161-TH: Advanced C Programming

**Lectures: 30 (Credits-2)**

**Prerequisites:**

- Basic knowledge of C language.

**Course Objectives:**

1. To study advanced concepts of programming using the C language.
2. To understand code organization with complex data types and structures.
3. To work with files.

**Learning Outcomes: On completion of this course, students will be able to:**

1. Develop modular programs using control structures, pointers, arrays, strings and structures.
2. Design and develop solutions to real world problems using C.

**Unit 1: Pointer** **08**

1.1 Introduction to Pointer
1.2 Declaration, definition, initialization, dereferencing
1.3 Pointer arithmetic
1.4 Relationship between Arrays & Pointers- Pointer to array, Array of pointers.
1.5 Multiple indirection (pointer to pointer)
1.6 Functions and pointers- Passing pointer to function, returning pointer from function, function pointer
1.7 Dynamic memory management- Allocation(malloc(), calloc()), Resizing (realloc()), Releasing(free()), Memory leak, dangling pointers

**Unit 2: String** **05**

2.1 Definition, declaration, initialization
2.2 Predefined string functions
2.3 Array of strings
2.4 Strings and Pointers
2.5 Command line arguments

**Unit 3: Structures and Unions** **07**

3.1 Concept of structure, definition and initialization, use of typedef
3.2 Accessing structure members
3.3 Nested Structures
3.4 Arrays of Structures
3.5 Structures and functions- Passing each member of structure as a separate argument, passing structure by value / address
3.6 Pointers and structures.
3.7 Self-referential structure and its application
3.8 Concept of Union, declaration, definition, accessing union members
3.9 Difference between structures and union

**Unit 4: File Handling and Preprocessor** **10**

4.1 Introduction to streams
4.2 Types of files
4.3 Operations on text files
4.4 Standard library input/output functions
4.5 Random access to files
4.6 Role of Preprocessor
4.7 Format of preprocessor directive
4.8 File inclusion directives (#include)

4.9 Macro substitution directive, argumented and nested macro
4.10 Macros versus functions

**Reference Books**


1. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill
2. The 'C' programming language, Brian Kernighan, Dennis Ritchie, PHI
3. Programming in C ,A Practical Approach, Ajay Mittal, Pearson
4. Problem Solving and Programming Concept, Maureen Sprankle, 7th Edition, Pearson Publication